1. Passport Automation System

1.1 Introduction

Getting a passport traditionally meant jumping through endless hoops—filling out paper forms, queuing for hours, visiting multiple government offices, and waiting in the dark to find out if your application was moving forward. On the flip side, passport officials had to wade through stacks of physical documents, manual identity checks, and paper-based tracking, which naturally caused long backlogs and occasional errors.

A modern Passport Automation System replaces that headache with a streamlined, digital process. By bringing every step online—from the moment an applicant submits their details and schedules an appointment, to background checks, printing, and final delivery—the platform eliminates unnecessary visits, protects sensitive identity data, speeds up processing times, and keeps citizens updated every step of the way.

1.2 Software Requirements Specification (SRS)

1.2.1 Purpose of this Document

This SRS document outlines the functional, technical, and operational requirements for building the Passport Automation System. It acts as a detailed blueprint for software developers, project managers, government stakeholders, law enforcement agencies, and passport office staff, ensuring everyone shares a clear understanding of the project's goals, scope, and deliverables.

1.2.2 Scope of the System

The system handles the entire lifecycle of a passport application. This includes new passport requests, renewals, re-issuances (for damaged or lost booklets), appointment booking, secure fee collection, police verification routing, administrative approval, booklet printing, and postal dispatch tracking.

- **Estimated Development Cost:** $100,000
- **Target Delivery Timeline:** 9 Months

1.2.3 System Overview

The platform acts as a secure central bridge connecting citizens, regional passport centers, police verification departments, and postal services. By replacing manual paperwork with automated workflows, it keeps data accurate, secure, and easily accessible to authorized personnel. Key modules include:

- **Citizen Self-Service Portal**
- **Document Management & AI Validation Engine**
- **Appointment & Capacity Management**
- **Payment Gateway Integration**
- **Law Enforcement & Verification Portal**
- **Adjudication & Approval Engine**
- **Printing & Postal Dispatch Tracking**
- **Analytics & Executive Dashboard**

1.2.4 General Description

Core Functions

- **Citizen Services:** Online registration, application filling (fresh issue, renewal, lost/damaged), identity document upload, fee payment, and real-time appointment booking.
- **Service Center Workflows:** On-site biometric capture (fingerprints/photo), document verification, and queue management at regional passport centers.
- **Verification Routing:** Automated transmission of applicant data to local police stations for field verification and criminal background checks.
- **Approval & Printing:** Review tools for Passport Officers to grant or reject applications, followed by automated dispatch of approved files to high-security printing centers.
- **Tracking & Updates:** Automated SMS, email, and portal notifications informing applicants as their passport moves through each stage.

User Objectives

- **Applicants:** A simple, transparent application process with clear instructions, online appointment slots, and real-time status updates.
- **Passport Center Staff:** Quick document verification tools and structured queues to handle daily walk-ins efficiently.
- **Police Verification Officers:** A secure digital portal to review background checks, upload field reports, and submit recommendations without paper files.
- **Government Leadership:** Real-time visibility into processing speeds, regional backlogs, revenue streams, and fraud detection metrics.

System Users

- **Citizens / Applicants:** Individuals applying for a new passport, renewal, or replacement.
- **Verification & Center Officers:** Staff at regional passport offices responsible for document checks, biometric capture, and final approvals.
- **Police & Law Enforcement Staff:** Local police officers assigned to complete field visits and background verification checks.
- **System Administrators:** IT staff managing user accounts, access permissions, server uptime, and security audits.

Key Benefits

- Faster turnaround times with fewer in-person visits to government offices.
- Full visibility for applicants through step-by-step progress updates.
- Reduced identity fraud through biometric verification and digital document validation.
- Data-driven reporting to help managers identify and fix operational bottlenecks.

1.2.5 Functional Requirements

- **Applicant Portal & Application Engine**
    - Support registration, secure login, multi-factor authentication, and draft application saving.

    - Dedicated application flows for **Fresh Passport**, **Renewal/Re-issue**, **Minor Passport**, and **Replacement (Lost/Damaged)**.
    - Automated form validation to prevent missing or invalid fields prior to submission.
- **Document & Biometric Management**
    - File uploads for identity, proof of address, date of birth, and passport-size photographs.
    - System tools allowing verification officers to review, mark as verified, flag discrepancies, or request document re-upload.
    - Integration with on-site biometric hardware for capturing digital fingerprints, facial photos, and signatures during office visits.
- **Smart Appointment Scheduling**
    - Real-time view of available time slots across regional Passport Centers.
    - Booking, rescheduling, and cancellation options with automated confirmation slips and QR-code entry passes.
- **Payment & Receipt Processing**
    - Auto-calculation of application fees based on application category, booklet size (36 vs. 60 pages), and processing speed (Standard vs. Expedited/Tatkaal).
    - Multiple payment options (credit/debit cards, net banking, digital wallets) with instant payment receipts.
- **Police Verification Module**
    - Automated routing of applicant data to the assigned local police jurisdiction based on residential address.
    - Digital portal for police officers to log field reports, mark status (**Clear / Adverse / Incomplete**), and attach supporting notes.
- **Adjudication & Final Approval**
    - Central dashboard for Passport Officers to review verified documents, biometric checks, and police reports in one place.
    - Single-click actions to **Approve**, **Reject (with official reason)**, or **Hold for Clarification**.
- **Printing, Dispatch & Live Tracking**
    - Automated queueing of approved applications for Machine-Readable Travel Document (MRTD) booklet printing.
    - Integration with national postal services to generate shipping labels and live tracking numbers.
    - Automated notification alerts sent via SMS, email, and WhatsApp/Mobile App.
- **Management Analytics & Reporting**
    - Daily, weekly, and monthly reports on total applications received, average processing times, pending verification backlogs, revenue collected, and rejection rates.

1.2.6 Interface Requirements

- **Software Interfaces**
    - **National Identity Database APIs:** Integration with government identity repositories for real-time verification of basic demographic details.
    - **Payment Gateways:** Secure integration with licensed payment aggregators.
    - **Police & Law Enforcement Portal API:** Encrypted channel linking passport servers with police verification databases.

    - **Postal & Logistics API:** API integration with national postal services for tracking booklet dispatches.
    - **Communication Gateways:** Bulk SMS, email, and push notification services.
- **User Interfaces**
    - **Citizen Portal & Mobile App:** Clean, high-accessibility UI built for web and mobile browsers.
    - **Officer Operations Dashboard:** High-density, efficient desktop dashboard for fast document review and approval workflows.
    - **Police Field Mobile App / Portal:** Lightweight interface optimized for tablets and smartphones used during field visits.

1.2.7 Performance Requirements

- **System Capacity:** Capable of supporting **15,000 concurrent user sessions** without speed drops.
- **Transaction Speed:** Form submissions and payment authorizations process in under **2.5 seconds**.
- **System Availability: 99.9% operational uptime** with automated failover systems.
- **Processing Accuracy:** Data entry error rate maintained below **0.05%**.

1.2.8 Technical & Design Constraints

- **Hardware & Infrastructure Constraints**
    - Cloud/On-Premise hybrid infrastructure with a primary data center (minimum 32 GB RAM, 2 TB SSD storage) and an active-passive **Disaster Recovery (DR)** site.
    - Standardized hardware support at regional centers for fingerprint scanners, webcam cameras, and document scanners.
- **Software Constraints**
    - Core system built using high-performance, secure frameworks (**Java / Spring Boot** or **Python / Django**).
    - Relational database backend using **PostgreSQL** or **Oracle** with full ACID compliance for secure ledger storage.
- **Legal & Regulatory Compliance**
    - Strict alignment with the **National Passports Act** and municipal passport issuance rules.
    - Full compliance with **ICAO Document 9303** standards for Machine-Readable Travel Documents (MRTD) and biometric data storage.
    - Adherence to national data privacy laws, data residency mandates, and government cyber-security frameworks.

1.2.9 Non-Functional Quality Attributes

- **Security:** Fine-grained Role-Based Access Control (RBAC), end-to-end data encryption (AES-256 for data at rest, TLS 1.3 for data in transit), and immutable audit logs tracking every administrative action.
- **Accessibility & Portability:** Web interface compliant with WCAG 2.1 AA accessibility guidelines, compatible across all major operating systems and modern mobile browsers.

- **Reliability & Resilience:** Daily automated database backups, zero-downtime deployment pipelines, and automatic failover in case of hardware failure.
- **Scalability:** Modular microservices architecture designed to scale seamlessly during peak application windows or policy expansions.
- **Data Integrity:** Database-level transactional safeguards to prevent record duplication or tampering.

1.2.10 Project Schedule and Budget

Project Timeline (9 Months Total)

1. **Discovery & Requirements Analysis:** 1 Month
2. **Architecture, Security Framework & UI Design:** 2 Months
3. **Core Development & Microservices Integration:** 5 Months
4. **Security Audits, User Acceptance Testing (UAT) & Launch:** 1 Month

Budget Allocation

- **Core Application Development:** $60,000
- **Quality Assurance, Security Audits & UAT:** $20,000
- **Deployment, DR Setup & Initial Maintenance:** $20,000
- **Total Estimated Budget: $100,000**

1.3 Class Diagram:

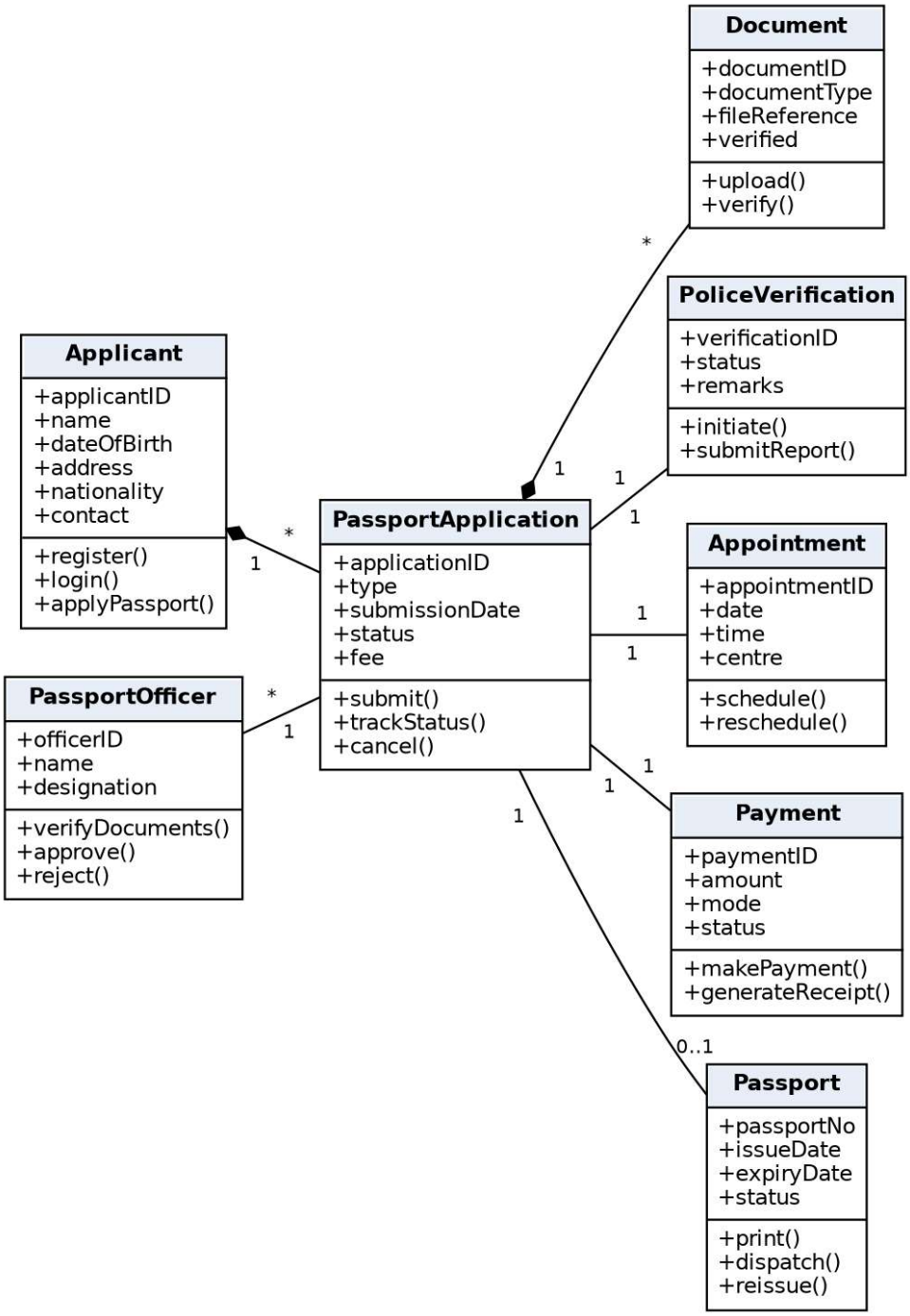


Fig 1.1

Brief Description:

The class diagram for the Passport Automation System (PAS) represents the key entities, their attributes, and relationships in the system. Below are the main components typically included in the class diagram for PAS:

- **Applicant:** stores citizen details and can submit one or more passport applications.

- **PassportApplication:** the central class holding application type, status and fee; it is linked to documents, appointment, payment, verification and the resulting passport.
- **Document:** represents an uploaded identity, address or photograph document and its verification status.
- **PassportOfficer:** verifies documents and approves or rejects applications.
- **PoliceVerification:** records the result of police verification for an application.
- **Appointment:** stores the date, time and centre booked by the applicant.
- **Payment:** records the application fee payment and its status.
- **Passport:** represents the issued passport with number, issue date, expiry date and status.